KAIST

2026 Special Topics in Physics Quantum Electronic Devices I:

Interferometers and Fractional Particles

Homework 2

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(1a)

$$U_{\mathrm{BS2}}U_{\mathrm{arm}}U_{\mathrm{BS1}}\left|S_1\right\rangle = \frac{1}{\sqrt{2}}U_{\mathrm{BS2}}U_{\mathrm{arm}}\left(\left|A\right\rangle + \left|A'\right\rangle\right) \tag{1}$$

$$= \frac{1}{\sqrt{2}}U_{\mathrm{BS2}}\left(c_1\left|A\right\rangle + c_2\left|A'\right\rangle\right) \tag{2}$$

$$= \frac{1}{2}(c_1+c_2)\left|D_1\right\rangle + \frac{1}{2}(c_1-c_2)\left|D_2\right\rangle \tag{3}$$

$$\sim \cos\theta\left|D_1\right\rangle + i\sin\theta\left|D_2\right\rangle \tag{4}$$

Where $\theta = k\Delta = k(L_1 - L_2)$.

The probability of detecting the particle at D_1 is given by

$$p_{D_1} = \frac{1}{4}|c_1+c_2|^2 = |\cos\theta|^2 \tag{5}$$

Hence,

$$\max[p_{D_1}] = 1, \quad \min[p_{D_1}] = 0 \tag{6}$$

and the visibility of the oscillation is

$$V = \frac{\max[p_{D_1}] - \min[p_{D_1}]}{\max[p_{D_1}] + \min[p_{D_1}]} = 1 \tag{7}$$

(1b)

Inculding the effect of the environment, we have

$$U_{\mathrm{BS2}}U_{\mathrm{arm}}^{SE}U_{\mathrm{BS1}}\left|S_1\right\rangle\left|0_{\mathrm{env}}\right\rangle = \frac{1}{\sqrt{2}}U_{\mathrm{BS2}}U_{\mathrm{arm}}^{SE}\left(\left|A\right\rangle\left|0_{\mathrm{env}}\right\rangle + \left|A'\right\rangle\left|0_{\mathrm{env}}\right\rangle\right) \tag{8}$$

$$= \frac{1}{\sqrt{2}}U_{\mathrm{BS2}}\left(c_1\cos\varphi\left|A\right\rangle\left|0_{\mathrm{env}}\right\rangle + c_1\sin\varphi\left|A\right\rangle\left|1_{\mathrm{env}}\right\rangle + c_2\left|A'\right\rangle\left|0_{\mathrm{env}}\right\rangle\right) \tag{9}$$

$$= \frac{1}{2}\left|D_1\right\rangle\left((c_1\cos\varphi + c_2)\left|0_{\mathrm{env}}\right\rangle + c_1\sin\varphi\left|1_{\mathrm{env}}\right\rangle\right) \tag{10}$$

$$+ \frac{1}{2}\left|D_2\right\rangle\left((c_1\cos\varphi - c_2)\left|0_{\mathrm{env}}\right\rangle + c_1\sin\varphi\left|1_{\mathrm{env}}\right\rangle\right) \tag{11}$$

The probability of detecting the particle at D_1 is changed due to the entanglement with the environment, and is given by

$$p_{D_1} = \frac{1}{4}\left|c_1\cos\varphi + c_2\right|^2 + \frac{1}{4}|c_1|^2\sin^2\varphi \tag{12}$$

$$= \frac{1}{2} + \frac{1}{2}\cos k\Delta\cos\varphi \tag{13}$$

The maximum and minimum values of p_{D_1} are

$$\max[p_{D_1}] = \frac{1+|\cos\varphi|}{2}, \quad \min[p_{D_1}] = \frac{1-|\cos\varphi|}{2} \tag{14}$$

and the visibility of the oscillation is

$$V = \frac{\max[p_{D_1}] - \min[p_{D_1}]}{\max[p_{D_1}] + \min[p_{D_1}]} = |\cos\varphi| \tag{15}$$

(1c)

The expectation of the projector on $|1_{\text{env}}\rangle$ is

$$\langle\psi|1_{\text{env}}\rangle\langle 1_{\text{env}}|\psi\rangle = \left\| \frac{1}{2} c_1 \sin\varphi \, |D_1\rangle + \frac{1}{2} c_1 \sin\varphi \, |D_2\rangle \right\|^2 = \frac{\sin^2\varphi}{2} \tag{16}$$

High $\sin^2\varphi$ means that the entanglement between the system and the environment is strong, and the which-path information is more accessible. Since $\sin^2\varphi + \cos^2\varphi = 1$, there is negative correlation between the which-path information and the visibility of the interference pattern.